1. Read, decode and make sense of statements... to create a model of the situation (Point 4)

- **Analysis:** This is the primary step in solving the problem. Students must decode the text information (one full rotation takes 40 minutes) and the image information (counter-clockwise rotation direction, starting point P, and various location points Q, R, S). Using this information, students need to build a mental model of the **correspondence between time and position**.

2. Recognise mathematical structure (regularities, relationships, and patterns) (Point 5)

- **Analysis:** The core of the problem lies in identifying the **periodic structure**. Students need to discover the proportional relationship between the rotation time and the position on the circumference. For example, realizing that 30 minutes is $\frac{3}{4}$ of the 40-minute cycle, or that every 10 minutes corresponds to $\frac{1}{4}$ of the circle's circumference. This recognition of patterns and regularities is the mathematical logical basis for solving the problem.

3. Identify the key variables in a model (Point 2)

- **Analysis:** Students need to extract the key variables from the description: the period ($T = 40\text{ min}$), the starting point (P), the direction of rotation (counter-clockwise), and the elapsed time ($t = 30\text{ min}$). Ignoring irrelevant variables (such as the height or diameter of the Ferris wheel) is crucial for simplifying the model.

4. Simplify or decompose a situation or problem... (Point 7)

- **Analysis:** To analyze the position after 30 minutes, students typically decompose the circular motion. For example, breaking down 30 minutes into $10 + 10 + 10$ minutes, corresponding to the movement from $P \rightarrow Q$ (10 min), $Q \rightarrow R$ (10 min), and $R \rightarrow S$ (10 min), respectively. This ability to break a complex process into manageable steps is fundamental to modeling.

5. Translate a problem into a standard mathematical representation... (Point 9)

- **Analysis:** This is a process of translating a real-world "amusement park ride operation" into "angle calculation" or "fractional operations". Students might represent it as $\frac{30}{40} = \frac{3}{4}$ of a circle rotation, or calculate the angle of rotation as $360^\circ \times 0.75 = 270^\circ$. This translation is a necessary step in using mathematical tools to solve the problem.

6. Recognise aspects of a problem that correspond with known problems... (Point 8)

- **Analysis:** This problem corresponds to common mathematical "clock problems" or "uniform circular motion" scenarios. Students utilize their existing procedural knowledge of "distance = speed × time" or "proportional distribution" to determine John's final coordinate point S on the circumference.

1. Read, decode and make sense of statements... to create a model of the situation (点 4)

- **分析：** 这是解题的首要步骤。学生必须解码文本信息（转一圈需要 40 分钟）和图像信息（逆时针旋转方向、起点 P、各个位置点 Q, R, S）。通过这些信息，学生需要在大脑中构建一个关于**时间与位置对应关系**的模型 (p. 1)。

2. Recognise mathematical structure (regularities, relationships, and patterns) (点 5)

- **分析：** 题目核心在于识别**周期性结构**。学生需要发现旋转时间与圆周位置之间的比例关系。例如，意识到 30 分钟是 40 分钟周期的 $\frac{3}{4}$，或者每 10 分钟经过 $\frac{1}{4}$ 圆周。这种对规律和模式的识别是解决该题的数学逻辑基础 (p. 1)。

3. Identify the key variables in a model (点 2)

- **分析：** 学生需要从描述中提取关键变量：周期 ($T = 40$ min)、起始点 (P)、旋转方向（逆时针）以及经过的时间 ($t = 30$ min)。忽略不相关的变量（如摩天轮的高度或直径）对于简化模型至关重要 (p. 1)。

4. Simplify or decompose a situation or problem... (点 7)

- **分析：** 为了分析 30 分钟后的位置，学生通常会将圆周运动分解。例如，将 30 分钟分解为 $10 + 10 + 10$ 分钟，分别对应从 $P \to Q$ (10min), $Q \to R$ (10min), $R \to S$ (10min)。这种将复杂过程分解为可分析步骤的能力是建模的核心 (p. 1)。

5. Translate a problem into a standard mathematical representation... (点 9)

- **分析：** 这是一个将现实中的“游乐设施运行”转化为“角度计算”或“分数运算”的过程。学生可能会将其表示为 $\frac{30}{40} = \frac{3}{4}$ 圈，或者计算旋转角度为 $360^\circ \times 0.75 = 270^\circ$。这种转化是利用数学工具解决问题的必经之路 (p. 1)。

6. Recognise aspects of a problem that correspond with known problems... (点 8)

- **分析：** 该题目对应于数学中常见的“时钟问题”或“圆周匀速运动”。学生利用已知的“路程 = 速度 × 时间”或“比例分配”的程序性知识来确定约翰在圆周上的最终坐标点 S (p. 1)。